# Post-cap pivots of the signed source

## P2 as alternation, P1 as a single prefix defect, and the aliasing remainder that is not a sign theorem

Stefan Hamann

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**Abstract**

On an odd cosine grid of $S = 2N - 1$ nodes the two Haynsworth premises of the dual two-rank cut translate, by a finite Hankel dictionary, into statements about the monic Jacobi norms $h_j = H_{j+1}(w)/H_j(w)$ of the original signed weight $w$:

$$\det K_2 < 0 \iff h_N(w)\, h_{N+1}(w) < 0, \qquad \operatorname{ind}_- H_{N+2}(w) \le 1.$$

The first equivalence is a ratio of Hankel determinants, checked in $\mathbb{Q}$ on a five-atom toy ($\det K_2 = -196/35719$ on both routes). It does *not* hold as a law of every canonical window: 35/85 MAIN rows have $h_N h_{N+1} > 0$, with the first negative pivot at offset 2, 3, or later (explicit counterexample $kz = 12$, first defect at $N + 2$, $\det K_2 = +12.46$, $A_0$ already positive definite). On the complementary nneg-1 branch the product is negative on 50/50 MAIN rows, and the unique prefix defect sits in $\{h_N, h_{N+1}\}$. The node polynomial $P_X(x) = 2^{1-S}(x+1)U_{S-1}(x)$ and the Chebyshev aliasing $T_{S+k}(x_j) = (-1)^j T_k(x_j)$ are theorems; they do not yield the sign of $h_N h_{N+1}$ without a free-window Gram inverse. The weak inertia bound $\operatorname{ind}_- H_{N+2} \le 1$ holds on 85/85 MAIN and 84/84 character-twisted windows, and fails on a matched scramble (first defect at degree 21, 37 negatives in the prefix). Nothing in this document is evidence for or against the Riemann hypothesis.

**Status.** Lemmas 1–6 are proved (finite algebra and trigonometric identities). Lemma 4 is proved (a $2 \times 2$ adversary). Proposition 1 is a named counterexample, not a cofinal law. Proposition 2 is a finite machine census of 85 MAIN and 84 character-twisted windows. Goal A as stated — $h_N h_{N+1} < 0$ on every canonical window — is *refuted.* Goal B — $\operatorname{ind}_- H_{N+2}(w) \le 1$ — is reduced to a construction census: it holds on every tested cosine-tent window and fails on scramble; a source Sturm uniqueness theorem was not obtained. No RH claim.

## 1 The object

Let $X = \{x_j = \cos(\pi j/S) : j = 1, \dots, S\}$ with $S = 2N - 1$ odd, and let $w$ be a real atomic weight on $X$ (the folded signed von-Mangoldt tent density times the mesh factor $1 - x$). Write $H_r(\rho)$ for the $r \times r$ Hankel determinant of a weight $\rho$ on $X$, $H_0 := 1$, and

$$h_j(\rho) = \frac{H_{j+1}(\rho)}{H_j(\rho)}$$

for the monic Jacobi norms whenever the minors are nonzero. The dual positive weight is $a(x) = 1/(|w(x)|P_X'(x)^2)$ and $\sigma a$ is its sign-dressed twin, $\sigma = \operatorname{sign} w$.

On the dual CD restriction $R_r$ to the negative atoms $Y$, the Haynsworth two-rank cut at $r = N - 3$ has premises

(P1) $\operatorname{ind}_-\big(R_{N-3} - \tfrac{1}{2}I\big) = 1$ (exactly one negative direction of $A_0$), (P2) $\det K_2 < 0$ with $K_2 = I_2 + U^T A_0^{-1} U$.

A weaker form of the first premise, used below as Goal B, is $\operatorname{ind}_-\big(R_{N-3} - \frac{1}{2}I\big) \le 1$. When the index is zero, $A_0$ is already positive definite and the Loewner update gives $R_{N-1} \succ \frac{1}{2}I$ without (P2).

## 2 The dictionary

**Lemma 1** (DPP and complement)**.** *In the a-orthonormal polynomial basis,* $\det(I - 2R_r) = H_r(\sigma a)/H_r(a)$. *The signed Borodin identity* $H_r(\sigma a) = H_{S-r}(w)/(\Delta(X)^2 \prod w)$ *holds for every rank* $r$*. Consequently, at* $r = N-3$*,*

$$\det K_2 = \frac{H_N(w)\, H_{N-3}(a)}{H_{N+2}(w)\, H_{N-1}(a)} = \frac{1}{h_N(w)\, h_{N+1}(w)\, h_{N-3}(a)\, h_{N-2}(a)}. \tag{1}$$

*Proof.* The first identity is Sylvester plus the change to the orthonormal basis of $a$ (the discrete OPE parity expectation). The second is Cauchy–Binet on the Vandermonde embedding of a signed atomic measure, the finite-set dual of [1]. Substitute $S = 2N-1$ and $r = N-3$, and rewrite consecutive Hankel ratios as $h_j$. □

**Lemma 2** (Sign dictionary)**.** *If* $a > 0$ *(all dual Hankels positive) and the named minors are nonzero, then* $\det K_2 < 0$ *if and only if* $h_N(w)\, h_{N+1}(w) < 0$.

*Proof.* Immediate from (1). □

**Lemma 3** (Complementary inertia)**.** $\operatorname{In} H_k(w) + \operatorname{In} H_{S-k}(\sigma a) = \operatorname{In} \operatorname{diag}(w)$ *for* $1 \le k \le S-1$. *For* $r \ge |Y|$, $\operatorname{ind}_-(R_r - \frac{1}{2}I) = \operatorname{ind}_- H_{S-r}(w)$. *In particular (P1) at rank* $N-3$ *is equivalent to* $\operatorname{ind}_- H_{N+2}(w) \le 1$ *in the weak form, and to equality with* 1 *on the nneg-*1 *branch.*

*Proof.* Vandermonde congruence plus a Schur complement whose inverse block is, after an invertible triangular change of basis, the complementary dual Hankel. The identification with $R_r - \frac{1}{2}I$ is the SVD of the rank-$r$ projector on $Y$. □

**Lemma 4** (Nonzero is not enough)**.** *There exist* $A = \operatorname{diag}(-2, 1)$ *and a two-column* $U$ *with* $\operatorname{ind}_- A = 1$ *and* $\det K_2 = \frac{1}{2} \ne 0$ *for which* $A + UU^T$ *remains indefinite.*

*Proof.* Take $U = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$. Then $K_2 = \operatorname{diag}(\frac{1}{2}, 1)$ and $M = \operatorname{diag}(-1, 1)$. □

On the nneg-1 branch, $\det K_2 < 0$ is equivalent to $M \succ 0$ by the determinant lemma $\det M = \det A \det K_2$ together with Loewner ($\operatorname{ind}_- M \le 1$) and the parity of a nonzero determinant. That circularity is real; it is not a collapse of (P2) onto (P1) plus nonvanishing (Lemma 4).

## 3 Aliasing on the cosine grid

**Lemma 5** (Node polynomial)**.** *The monic node polynomial of* $X$ *is* $P_X(x) = 2^{1-S}(x+1)U_{S-1}(x)$. *It vanishes on every node, including* $x = -1 = \cos\pi$.

*Proof.* $U_{S-1}(\cos\theta) = \sin(S\theta)/\sin\theta$ has simple zeros at $\theta = \pi j/S$ for $j = 1, \ldots, S-1$. The remaining node $\theta = \pi$ is the factor $x+1$. The leading coefficient of $U_{S-1}$ is $2^{S-1}$, so $2^{1-S}(x+1)U_{S-1}$ is monic of degree $S$. □

**Lemma 6** (Chebyshev aliasing)**.** *On* $X$, $T_{S+k}(x_j) = (-1)^j T_k(x_j)$ *for every* $k \ge 0$. *In particular* $T_S(x_j) = (-1)^j$.

*Proof.* $T_n(\cos\theta) = \cos(n\theta)$ and $\theta_j = \pi j/S$, so $\cos((S+k)\theta_j) = \cos(\pi j + k\theta_j) = (-1)^j \cos(k\theta_j)$. □

The power-moment sequence of any atomic measure on $X$ is linearly recurrent of order $S$ with characteristic polynomial $P_X$. The first two post-cap norms $h_N, h_{N+1}$ are exactly the first two Jacobi steps that consume recurrent (aliased) moments: $H_N$ uses the complete independent window $m_0, \ldots, m_{S-1}$, while $H_{N+1}$ and $H_{N+2}$ use $m_S, \ldots, m_{S+3}$. The classical bordered identity

$$h_N = m_{2N} - v^T H_N^{-1} v$$

is this step in the monomial basis. It is a free-window Gram inverse — the kill criterion of the aliasing route. Live power-moment recurrences at $S = 367$ cancel in IEEE-754; the trigonometric form of Lemma 5 is exact ($\max|P_X| \le 10^{-123}$ on the construction grid).

A low-mode density $f = 1 + 0.3\cos\theta + 0.1\cos 2\theta$ on $S = 5$ produces a strictly positive weight times $(1-x)$ and therefore $h_j > 0$ for all $j$: post-cap alternation is not a theorem of the cosine mesh alone. The sign, when it occurs, is carried by the high Fourier content of the tent comb. No identity was found that reads that sign from the two Nyquist DCT blocks without inverting the free-window Gram.

# 4 Goal A, refuted as stated

**Proposition 1** (Universal post-cap alternation is false)**.** *There exist canonical windows with $h_N(w)\,h_{N+1}(w) > 0$. On $kz = 12$ one has $N = 151$, $h_0, \ldots, h_{152} > 0$, $h_{153} < 0$, $\det K_2 = +12.4632$, and $A_0 \succ 0$.*

The same pattern — first defect at offset 2, 3, 4, 5, or 8 — occurs on 35/85 MAIN rows (classes LATE and NONE of the census). The twelve NONE rows, hunted to $N + 16$, have offsets $\{4, 5, 8\}$: $kz \in \{50, 73, 96, 72, 113\}$ at 4; $kz \in \{43, 42, 115, 89, 133\}$ at 5; $kz \in \{107, 117\}$ at 8. It is compatible with half-filling positivity $h_0, \ldots, h_{N-1} > 0$, which holds on 85/85 MAIN and 84/84 character-twisted rows. The eighteen $\delta = 0$ core windows with $h_N < 0$ (r241) are exactly the PINNED-$N$ subclass of the core 42.

On the nneg-1 branch ($\mathrm{ind}_- H_{N+2} = 1$) the unique prefix defect lies in $\{h_N, h_{N+1}\}$ and the product is negative, on 50/50 MAIN rows (19 at $N$, 31 at $N + 1$). That is the content of (P2) after Lemma 2. It is a finite census, not a source sign theorem: the aliasing identities of §3 do not evaluate the product.

# 5 Goal B, reduced to a construction census

A second negative pivot in $h_0, \ldots, h_{N+1}$ would be a second sign change of the Sturm sequence of monic norms (LDL of the signed Hankel). The three-term recurrence $\pi_{n+1} = (x - \alpha_n)\pi_n - \gamma_n \pi_{n-1}$ with $\gamma_n = h_n/h_{n-1}$ makes every sign of $h$ a recurrence coefficient, but does not by itself forbid a second change: a generic signed atomic measure (the five-atom toy, or a matched scramble) has many early negatives.

**Proposition 2** (Prefix census)**.** *On the* 85 *MAIN windows,* $\mathrm{ind}_- H_{N+2}(w) \le 1$ *holds* 85/85*, with* 50 *rows equal to* 1 *and* 35 *equal to* 0 *(PINNED-$N$* 19*, PINNED-$N$+*1 31*, LATE* 23*, NONE* 12*; DOUBLE/EARLY/MULTI* 0*). On the* 84 *character-twisted windows the same bound holds* 84/84 *($\chi_3$: PINNED-$N$* 8*, PINNED-$N$+*1 13*, LATE* 18*, NONE* 3*; $\chi_4$:* 4, 15, 23, 0*). Defects on $\chi$ wander through offsets* 0..3 *(and three $\chi_3$ NONE). The six terminal-dead character rungs ($q_N > 1$) still satisfy the bound ($\chi_3$:* 15/19 *PINNED-$N$+*1*,* 23/33 *LATE,* 39 *PINNED-$N$; $\chi_4$-*20 *LATE); terminal death is not a prefix-pivot failure. A matched scramble at*

*$w = 9$ breaks the bound (first defect at degree 21, 37 negatives among $h_0, \ldots, h_{185}$). No row of the cosine-tent family realises two negatives in $\{h_N, h_{N+1}\}$.*

Rank-one Loewner monotonicity $\mathrm{ind}_-(A_{n+1}) \in \{\mathrm{ind}_-(A_n), \mathrm{ind}_-(A_n) - 1\}$ (Fable, Weyl plus interlacing) then spreads the bound from rank $N - 3$ through the tail. The missing source lemma is a reason that the cosine-tent density cannot produce a second prefix sign change. Saturation bulk coercivity ($x \perp \psi_{\mathrm{edge}} \Rightarrow x^T A_{n_0} x > 0$) was not needed: the census never entered a two-defect prefix, so there was no second mode to coerce.

## 6 Outcomes

**Goal A (P2-POSTCAP).** *Refuted* as the universal statement $h_N h_{N+1} < 0$ on every canonical window (Proposition 1, 35 MAIN counterexamples). *Reduced*, on the nneg-1 branch, to a 50/50 census of alternation at $\{h_N, h_{N+1}\}$, with the exact dictionary Lemma 2 and with aliasing stopped at the $H_N^{-1}$ kill.

**Goal B (P1-SINGLE-DEFECT).** *Reduced* to Proposition 2: $\mathrm{ind}_- H_{N+2} \leq 1$ on every tested cosine-tent window, named-broken by scramble, not proved from the Sturm recurrence. The bound is not zeta-specific (character-twisted and terminal-dead rungs obey it).

**What remains.** A source reason that the first aliased Jacobi step of a cosine-tent weight is a *simple* Nyquist crossing on the nneg-1 branch, without inverting $H_N$. The dictionary, the node polynomial, and the Chebyshev wrap are no longer the obstruction.

## References


[1] A. Borodin, *Duality of random discrete sets*, arXiv:math/0101125.